

CSA0501
DATABASE MANAGEMENT SYSTEMS
LAB EXPERIMENT

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EXP 9:

AIM

To perform JOIN using Equi Join, Inner Join, Outer Join on the given relation.

ALGORITHM

1. Start.
2. Create a database and select it.
3. Create the DEPARTMENT, FACULTY, COURSE, and STUDENT tables with appropriate columns.
4. Insert sample records into all four tables.
5. Use INNER JOIN to combine FACULTY and DEPARTMENT to find departments where faculty work.
6. Use INNER JOIN with a WHERE condition to find students with zero marks in their courses.
7. Use multiple INNER JOINs across FACULTY, DEPARTMENT, and COURSE to find courses taken by a specific faculty in a specific department.
8. Display and verify the output of each query.
9. Stop.

COMMANDS (short format)

sql

```
CREATE DATABASE dbms_lab9;
```

```
USE dbms_lab9;
```

```
CREATE TABLE department (...);
```

```
CREATE TABLE faculty (...);
```

```
CREATE TABLE course (...);
```

```
CREATE TABLE student (...);
```

```

INSERT INTO department VALUES (...);

INSERT INTO faculty VALUES (...);

INSERT INTO course VALUES (...);

INSERT INTO student VALUES (...);

SELECT faculty.facno, faculty.facname, department.deptno, department.deptname
FROM faculty INNER JOIN department ON faculty.deptno = department.deptno;

SELECT student.name, student.marks, course.courseno
FROM student INNER JOIN course ON student.courseno = course.courseno
WHERE student.marks = 0;

SELECT course.courseno, course.coursedesc
FROM faculty
INNER JOIN department ON faculty.deptno = department.deptno
INNER JOIN course ON faculty.facno = course.facno
WHERE department.deptname = 'Sales' AND faculty.facname = 'Ratnam';

```

QUESTIONS

1. List the departments where the faculty members are working.
2. Find the student who has no score in any of the courses. List student name and course number.
3. The office clerk needs the names of the courses taken by the faculty belonging to 'Sales' department whose name is 'Ratnam'.

OUTPUTS:

```

mysql> SELECT faculty.facno, faculty.facname, department.deptno, department.deptname
-> FROM faculty
-> INNER JOIN department ON faculty.deptno = department.deptno;
+-----+-----+-----+-----+
| facno | facname | deptno | deptname |
+-----+-----+-----+-----+
| F002  | Kavitha | D01    | CSE      |
| F003  | Suresh  | D02    | ECE      |
| F001  | Ratnam  | D03    | Sales    |
+-----+-----+-----+-----+
3 rows in set (0.04 sec)

```

```
mysql> SELECT student.name, student.marks, course.courseno
-> FROM student
-> INNER JOIN course ON student.courseno = course.courseno
-> WHERE student.marks = 0;
```

name	marks	courseno
Pooja	0	C01
Geetha	0	C01
Ramu	0	C01

3 rows in set (0.01 sec)

```
mysql> SELECT course.courseno, course.coursedesc
-> FROM faculty
-> INNER JOIN department ON faculty.deptno = department.deptno
-> INNER JOIN course ON faculty.facno = course.facno
-> WHERE department.deptname = 'Sales' AND faculty.facname = 'Ratnam';
```

courseno	coursedesc
C03	Sales Mgmt

1 row in set (0.01 sec)

RESULT

The records from the tables are displayed using JOIN — Equi Join, Inner Join, Outer Join.

Ready for the next experiment whenever you are — just share the manual pages.